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Digital image capturing devices including lens modules with integrated sensor-lens assemblies

Abstract

A digital image capturing device (DICD) that includes: a first integrated sensor-lens assembly (ISLA) facing in a first direction; a second ISLA facing in a second direction generally opposite to the first direction; and a bridge member that is positioned between the first ISLA and the second ISLA, wherein the bridge member is configured as a discrete structure that is separate from the first ISLA and the second ISLA.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/420,778, filed on Jul. 6, 2021, which is a 371 of International Application No. PCT/US2020/012022, filed on Jan. 2, 2020, which claims priority to U.S. Provisional Application No. 62/788,433, filed on Jan. 4, 2019, the entire contents of each of the above-identified applications being hereby incorporated by reference.

TECHNICAL FIELD

(1) The present disclosure relates generally to digital image capturing devices (DICDs), and, more specifically, to DICDs including dual integrated sensor-lens assemblies (ISLAs).

BACKGROUND

(2) DICDs, such as digital cameras, are used in various applications, including, for example, handheld cameras and video recorders, drones, and vehicles. DICDs capture, focus, and convert

light into an electronic image signal using an optical module that typically includes one or more lenses and image sensors. To produce a high-quality, focused image, however, each lens must be properly aligned with a corresponding image sensor in multiple degrees-of-freedom. Even small misalignments (e.g., in position or tilt of the lens and/or the image sensor) can negatively impact the image generated by the DICD.

(3) To address this concern, active alignment procedures have been developed that continually measure and adjust the positions of the lens and/or the image sensor during assembly of the optical module to optimize alignment. The present disclosure discusses various advancements in active alignment technology as it relates to spherical DICDs including multiple lenses and multiple image sensors.

SUMMARY

(4) In one aspect of the disclosure, an optical module for a digital image capturing device (DICD) is described. The optical module includes a first integrated sensor-lens assembly (ISLA), a second ISLA, and a bridge member configured as a discrete structure (i.e., as being separate from both the first ISLA and the second ISLA). The first ISLA defines a first optical axis and includes a first lens; a first image sensor that is fixed in relation to the first lens; and a first mount that is configured to support the first lens and the first image sensor. The second ISLA defines a second optical axis that is offset from the first optical axis and includes a second lens; a second image sensor that is fixed in relation to the second lens; and a second mount that is configured to support the second lens and the second image sensor. The first mount includes a first receptacle and the second mount includes a second receptacle that collectively defines a cavity with the first receptacle. The first and second ISLAs are identical in configuration and are oriented in generally opposite directions (i.e., such that the second ISLA is rotated approximately 180° from the first ISLA), and the bridge member is configured for receipt within the cavity defined by the first and second receptacles to thereby fixedly secure together the first and second ISLAs.

(5) In certain embodiments, the first ISLA may define a first field-of-view of approximately 180° to approximately 210° , and the second ISLA may define a second field-of-view of approximately 180° to approximately 210° . In such embodiments, the first and second ISLAs may be positioned such that the first and second fields-of-view overlap, whereby images captured by the first and second ISLAs can be stitched together to form a single image (e.g., a spherical image). In certain embodiments, the bridge member may include a first bridge portion, and a second bridge portion that is connected to the first bridge portion. In certain embodiments, the first and second bridge portions may be identical in configuration and positioned in generally opposite orientations (i.e., the second bridge portion may be rotated approximately 180° from the first bridge portion). In certain embodiments, the bridge member may define a first node that is located adjacent a first end of the bridge member; a second node that is located adjacent a second, opposite end of the bridge member; and a third node that is positioned between the first and second nodes. In certain embodiments, the cavity defined by the first and second receptacles may include a first cavity portion that is configured to receive the first node of the bridge member; a second cavity portion that is configured to receive the second node of the bridge member; and a third cavity portion that is configured to receive the third node of the bridge member. In certain embodiments, the first and second receptacles may each define a first end chamber, a second end chamber, and an intermediate chamber that is positioned between the first and second end chambers. In certain embodiments, the first end chamber of the first receptacle may be positioned generally opposite the second end chamber of the second receptacle so as to define the first cavity portion, the second end chamber of the first receptacle may be positioned generally opposite the first end chamber of the second receptacle so as to define the second cavity portion, and the intermediate chamber of the first receptacle may be positioned generally opposite the intermediate chamber of the second receptacle so as to define the third cavity portion. It is envisioned that the optical module described above may include any combination of the features and the elements described in this paragraph.

(6) In another aspect of the present disclosure, a digital image capturing device (DICD) is described that includes a first integrated sensor-lens assembly (ISLA) defining a first optical axis and facing in a first direction; a second ISLA defining a second optical axis offset from the first optical axis and facing in a second direction generally opposite the first direction (i.e., such that the second ISLA is rotated approximately 180° from the first ISLA); and a bridge member that is positioned between the first and second ISLAs to fixedly secure together the first and second ISLAs. The bridge member is configured as a discrete structure (i.e., as being separate from both the first ISLA and the second ISLA), and defines a longitudinal axis that is generally parallel in relation to the first and second optical axes.

(7) In certain embodiments, the first and second ISLAs may be identical in configuration. In certain embodiments, the first ISLA may define a first field-of-view of approximately 180° to approximately 210° , and the second ISLA may define a second field-of-view of approximately 180° to approximately 210° . In such embodiments, the first and second ISLAs may be positioned such that the first and second fields-of-view overlap, whereby images captured by the first and second ISLAs can be stitched together to form a single image (e.g., a spherical image). In certain embodiments, the first ISLA may include a first lens; a first image sensor fixed in relation to the first lens; and a first mount that is configured to support the first lens and the first image sensor. Correspondingly, the second ISLA may include a second lens; a second image sensor fixed in relation to the second lens; and a second mount that is configured to support the second lens and the second image sensor. In certain embodiments, the first mount may define a first receptacle, and the second mount may define a second receptacle, wherein the first and second receptacles are each configured to receive the bridge member to thereby connect the first ISLA and the second ISLA. In certain embodiments, the DICD may further include a curable adherent (e.g., an adhesive that is curable via exposure to UV light and/or heat) that is positioned between the bridge member and the first and second mounts to fixedly secure the bridge member to the first and second mounts. In certain embodiments, the bridge member may include a first bridge portion, and a second bridge portion that is connected to the first bridge portion. In certain embodiments, the first and second bridge portions may be identical in configuration and may each include a first end section; an opposite second end section; and an intermediate section positioned between the first and second end sections. The first and second bridge portions may be oriented in generally opposite directions (i.e., the second bridge portion may be rotated approximately 180° from the first bridge portion) such that the first end section of the first bridge portion is positioned adjacent the second end section of the second bridge portion so as to define a first node, the second end section of the first bridge portion is positioned adjacent the first end section of the second bridge portion so as to define a second node, and the intermediate section of the first bridge portion is positioned adjacent the intermediate section of the second bridge portion so as to define a third node that is positioned between the first and second nodes. In certain embodiments, the first and second receptacles may each define a first end chamber; a second end chamber; and an intermediate chamber that is positioned between the first and second end chambers, wherein the chambers of the first receptacle and the chambers of the second receptacle collectively define a cavity that is configured to receive the bridge member. In certain embodiments, the first and second ISLAs may be positioned such that: the first end chamber of the first receptacle is positioned generally opposite the second end chamber of the second receptacle so as to define a first cavity portion that is configured to receive the first node of the bridge member; the second end chamber of the first receptacle may be positioned generally opposite the first end chamber of the second receptacle so as to define a second cavity portion that is configured to receive the second node of the bridge member; and the intermediate chamber of the first receptacle may be positioned generally opposite the intermediate chamber of the second receptacle so as to define a third cavity portion that is configured to receive the third node of the bridge member. It is envisioned that the optical module described above may include any combination of the features and the elements described in this paragraph. It is

envisioned that the DICD described above may include any combination of the features and the elements described in this paragraph.

(8) In another aspect of the disclosure, a method is described for assembling an optical module for a digital image capturing device (DICD) that includes first and second identical integrated sensor-lens assemblies (ISLAs). The method includes actively aligning a lens and an image sensor of the first ISLA; actively aligning a lens and an image sensor of the second ISLA; connecting a bridge member to the first ISLA; connecting the second ISLA to the bridge member such that optical axes defined by the first and second ISLAs are offset from one another, and the first and second ISLAs are oriented in generally opposite directions (i.e., such that the first and second ISLAs are rotated approximately 180° from one another); and actively aligning the first and second ISLAs with respect to one another.

(9) In certain embodiments, connecting the bridge member to the first ISLA and connecting the second ISLA to the bridge member may include positioning the bridge member within a cavity collectively defined by the first ISLA and the second ISLA. In certain embodiments, positioning the bridge member within the cavity may include positioning the bridge member within a first receptacle defined by the first ISLA and within a second receptacle defined by the second ISLA. In certain embodiments, the cavity may include: a first cavity portion; a second cavity portion; and a third cavity portion. In such embodiments, positioning the bridge member may include positioning a first node of the bridge member within the first cavity portion, positioning a second node of the bridge member within the second cavity portion, and positioning a third node of the bridge member within the third cavity portion. In certain embodiments, connecting the bridge member to the first ISLA and connecting the second ISLA to the bridge member may include curing an adherent (e.g., an adhesive that is curable via exposure to UV light and/or heat) that is positioned between the bridge member and the receptacles defined by the first and second ISLAs. It is envisioned that the method described above may include any combination of the features and the elements described in this paragraph.

(10) In another aspect of the disclosure, a digital image capturing device (DICD), is disclosed that includes a first integrated sensor-lens assembly (ISLA) facing in a first direction and defining a first optical axis; a second ISLA facing in a second direction generally opposite to the first direction and defining a second optical axis that is offset from the first optical axis; and a bridge member that is positioned between the first ISLA and the second ISLA to fixedly secure together the first ISLA and the second ISLA. The bridge member is configured as a discrete structure that is separate from the first ISLA and the second ISLA and defines a longitudinal axis that is generally parallel in relation to the first optical axis and the second optical axis.

(11) In certain embodiments, the first ISLA may include a first lens; a first image sensor that is fixed in relation to the first lens; and a first mount that is configured to support the first lens and the first image sensor. In certain embodiments, the second ISLA may include a second lens; a second image sensor that is fixed in relation to the second lens; and a second mount that is configured to support the second lens and the second image sensor. In certain embodiment, the first mount may define a first receptacle and the second mount may define a second receptacle. In certain embodiments, the first receptacle and the second receptacle may each be configured to receive the bridge member to thereby connect the first ISLA and the second ISLA. In certain embodiments, the bridge member may include a first bridge portion and a second bridge portion that is connected to the first bridge portion. In certain embodiments, the first bridge portion and the second bridge portion may be identical in configuration and oriented in generally opposite directions. In certain embodiments, the first bridge portion and the second bridge portion may each include a first end section; a second end section opposite to the first end section; and an intermediate section that is positioned between the first end section and the second end section. In certain embodiments, the first end section of the first bridge portion may be positioned adjacent to the second end section of the second bridge portion so as to define a first node, the second end section of the first bridge

portion may be positioned adjacent to the first end section of the second bridge portion so as to define a second node, and the intermediate section of the first bridge portion may be positioned adjacent to the intermediate section of the second bridge portion so as to define a third node that is positioned between the first node and the second node. In certain embodiments, the first receptacle and the second receptacle may each define a first end chamber; a second end chamber; and an intermediate chamber that is positioned between the first end chamber and the second end chamber. In certain embodiments, the chambers of the first receptacle and the chambers of the second receptacle may collectively define a cavity that is configured to receive the bridge member. In certain embodiments, the first ISLA and the second ISLA may be positioned such that the first end chamber of the first receptacle is positioned generally opposite to the second end chamber of the second receptacle so as to define a first cavity portion that is configured to receive the first node of the bridge member, the second end chamber of the first receptacle is positioned generally opposite to the first end chamber of the second receptacle so as to define a second cavity portion that is configured to receive the second node of the bridge member, and the intermediate chamber of the first receptacle is positioned generally opposite to the intermediate chamber of the second receptacle so as to define a third cavity portion that is configured to receive the third node of the bridge member. In certain embodiments, the first ISLA and the second ISLA may be identical in configuration. In certain embodiments, the first ISLA may define a first field-of-view of approximately 180° to approximately 210° and the second ISLA may define a second field-of-view of approximately 180° to approximately 210°. In certain embodiments, the first ISLA and the second ISLA may be positioned such that the first field-of-view and the second field-of-view overlap, whereby images captured by the first ISLA and the second ISLA can be stitched together to form a single image. In certain embodiments, the DICD may further include a curable adherent that is positioned between the bridge member, the first ISLA, and the second ISLA to fixedly secure the bridge member to the first ISLA and to the second ISLA. It is envisioned that the DICD described above may include any combination of the features and the elements described in this paragraph.

(12) In another aspect of the present disclosure, a method of assembling an optical module for a digital image capturing device (DICD) including first and second identical integrated sensor-lens assemblies (ISLAs) is disclosed. The method includes actively aligning a lens and an image sensor of the first ISLA; actively aligning a lens and an image sensor of the second ISLA; connecting a bridge member to the first ISLA; connecting the second ISLA to the bridge member such that optical axes defined by the first ISLA and the second ISLA are offset from one another and the first ISLA and the second ISLA are oriented in generally opposite directions; and actively aligning the first ISLA and the second ISLA with respect to one another.

(13) In certain embodiments, connecting the bridge member to the first ISLA and connecting the second ISLA to the bridge member may include positioning the bridge member within a cavity collectively defined by the first ISLA and the second ISLA. In certain embodiments, positioning the bridge member within the cavity may include positioning the bridge member within a first receptacle defined by the first ISLA and within a second receptacle defined by the second ISLA. In certain embodiments, positioning the bridge member within the cavity may include positioning a first node of the bridge member within a first cavity portion; positioning a second node of the bridge member within a second cavity portion; and positioning a third node of the bridge member within a third cavity portion. In certain embodiments, connecting the bridge member to the first ISLA and connecting the second ISLA to the bridge member may include curing an adherent that is positioned between the bridge member, the first ISLA, and the second ISLA. It is envisioned that the method described above may include any combination of the features and the elements described in this paragraph.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings may not be to scale, and the dimensions of the various components may be arbitrarily expanded or reduced for clarity.

(2) FIG. 1A is a front, perspective view of a DICD including an optical module with identical (first and second) ISLAs oriented in generally opposite directions in accordance with the principles of the present disclosure.

(3) FIG. 1B is a rear, perspective view of the DICD seen in FIG. 1.

(4) FIG. 2 is a cross-sectional view of the DICD taken through lines 2-2 in FIGS. 1A and 1B.

(5) FIG. 3 is block diagram illustrating various internal components of the DICD.

(6) FIG. 4 is a top, perspective view of the optical module with parts separated illustrating connection of the ISLAs using a bridge member.

(7) FIG. 5A is a top, cross-sectional view of the optical module upon assembly.

(8) FIG. 5B is a side, perspective view of the optical module upon assembly.

(9) FIG. 6 is a top, perspective view of one of the ISLAs with parts separated.

(10) FIG. 7 is a bottom, perspective view of the presently disclosed bridge member.

(11) FIG. 8 is a top, perspective view of the presently disclosed bridge member.

(12) FIG. 9 is a partial, top, perspective view of the first ISLA.

(13) FIG. 10 is a partial, bottom, perspective view of the ISLA seen in FIG. 9.

(14) FIG. 11 is a partial, top, perspective view of the second ISLA.

(15) FIG. 12 is a top, perspective view of the optical module upon assembly.

(16) FIG. 13 is a flow chart illustrating a method of assembling the optical module.

(17) FIG. 14 is a cross-sectional view of the DICD taken through lines 2-2 in FIGS. 1A and 1B illustrating non-identical (first and second) ISLAs according to an alternate embodiment of the disclosure.

(18) FIG. 15A is a partial, top, perspective view of the first ISLA seen in FIG. 14.

(19) FIG. 15B is a partial, side, plan view of the first ISLA.

(20) FIG. 16A is a partial, top, perspective view of the second ISLA seen in FIG. 14.

(21) FIG. 16B is a partial, side, plan view of the second ISLA.

DETAILED DESCRIPTION

(22) The present disclosure describes a DICD including an optical module with dual (first and second) ISLAs, which may be identical or non-identical in configuration in various embodiments of the disclosure. The ISLAs are oriented in generally opposite directions (i.e., such that the ISLAs are rotated approximately 180° from one another) and include overlapping fields-of-view so as to support the capture and creation of spherical images. The ISLAs are fixedly connected to one another (e.g., by a bridge member), which allows for a reduction in not only the overall number of components in the optical module, but the complexity typically associated with the assembly of spherical DICDs. Many known spherical DICDs, for example, require a high number of mechanical connections and/or interfaces between ISLAs, each of which must be adjusted during the alignment process, often within very small tolerances (e.g., on the order of microns), which results in a lengthy, complex assembly process with a low yield. The structures and methods described herein address this issue by reducing the number of mechanical connections and/or interfaces between the ISLAs via attachment through the aforementioned bridge member.

(23) During assembly of the optical module, the ISLAs are separately subjected to an active alignment process, during which, optimal alignment is achieved between the lens and the image sensor included in each ISLA. Subsequently, the various components of the ISLAs are secured

together using UV and thermal curing processes, the components of the optical module (e.g., the ISLAs and the bridge member) are cleaned (e.g., via a plasma cleaning process) to prepare the components for connection, and the individual ISLAs are connected to each other.

(24) To connect the ISLAs, initially, an adherent is applied to the first ISLA and/or the bridge member, and a first apparatus (e.g., a robotic assembly) is used to position and align the bridge member with respect to the first ISLA. Once properly positioned, the adherent is subjected to UV and thermal curing processes to fix the connection between the first ISLA and the bridge member. After allowing the assembly of the first ISLA and the bridge member to cool, adherent is applied to the second ISLA and/or the bridge member, and the assembly of the first ISLA and the bridge member is connected to the second ISLA using a second apparatus (e.g., a robotic assembly), which positions and aligns the bridge member with respect to the second ISLA. The optical module is then subjected to an additional active (bore-sight) alignment process to align the ISLAs with respect to one another. More specifically, while viewing one or more optical targets, the positions of the ISLAs are continually adjusted until optimal alignment is achieved. Once properly aligned, the adherent is subjected to UV and thermal curing processes to fix the relative positions of the ISLAs and the bridge member, and thereby preserve alignment.

(25) With reference now to FIGS. 1A-3, an example DICD **100** is illustrated for use in capturing (and storing) digital data including, for example, images, video, and audio. The DICD **100** includes a body **102**, an optical module **104** (FIG. 2) with dual (e.g., first and second) ISLAs **106A**, **106B** that define respective fields-of-view **108A**, **108B**, and a bridge member **110** that is positioned between the ISLAs **106A**, **106B** to fixedly connect the ISLAs **106A**, **106B**.

(26) The body **102** supports the optical module **104**, as well as the various other internal and external components of the DICD **100**, and may include (e.g., may be formed from) any material or combination of materials suitable for this intended purpose, such as plastic, aluminum, steel, fiberglass, etc. For example, as seen in FIGS. 1A and 1B, the body **102** includes a variety of indicators, such as one or more LED lights **112** and an LED display **114**, as well as one or more buttons **116** that allow a user to interact with the DICD **100** (e.g., to turn the DICD **100** on and off, configure the operating mode, etc.). The body **102** may also support a microphone **118** to allow the DICD **100** to receive and/or record audio signals, an I/O interface **119** (e.g., to transmit to, and receive data from, various external devices), and various other input mechanisms (e.g., switches, touchscreen mechanisms, etc.). Although the body **102** of the DICD **100** is illustrated as including six external surfaces (i.e., a front face **102A**, a rear face **102B**, a left face **102C**, a right face **102D**, a top face **102E**, and a bottom face **102F**) defining a generally rectangular-cuboid configuration, the particular geometric configuration of the body **102** may be varied in alternate embodiments without departing from the scope of the present disclosure.

(27) Image data captured by the optical module **104** may be combined or merged together to produce a combined image. For example, by correlating overlapping regions in the respective fields-of-view **108A**, **108B** of the ISLAs **106A**, **106B**, and by stitching together the data captured by the ISLAs **106A**, **106B**, a cohesive, combined spherical or panoramic image may be generated. In certain implementations of the DICD **100**, it is envisioned that the generation of such an image may include additional supportive processes, such as three-dimensional (or spatiotemporal) noise reduction (3DNR) and/or matching pixels along the stitch boundary to reduce (or entirely eliminate) discontinuities.

(28) To facilitate image processing and general operability, the DICD **100** may also include a variety of additional components, such as one or more processors **120** (FIG. 3) (e.g., a camera system-on-chip), a communications interface **122** (e.g., for transferring images to other devices), a user interface **124**, and a power source (e.g., a battery) **126**, which may be charged in any suitable manner (e.g., wirelessly or through a micro-USB interface). It is envisioned that the various components of the DICD **100** may communicate with one other in any suitable manner, such as, for example, via a bus **128**.

(29) The processor(s) **120** are configured to receive a first image from the ISLA **106A** and a second image from the ISLA **106B**, and may be configured to perform image signal processing (e.g., filtering, tone mapping, stitching, and/or encoding) to generate output images based on the image data received from the ISLAs **106A**, **106B**. The processor(s) **120** may include one or more individual processors (with single or multiple processing cores), and may include any suitable memory, such as, for example, a random-access memory device (RAM), flash memory, or other such storage medium (e.g., a non-transitory computer-readable memory). It is envisioned that the memory may include executable instructions and/or data that is accessible by the processor(s) **120**. For example, the processor(s) **120** may include one or more dynamic random-access memory (DRAM) modules, such as double data rate synchronous dynamic random-access memory (DDR SDRAM), a digital signal processor (DSP), and/or an application-specific integrated circuit (ASIC) (e.g., a custom image signal processor).

(30) The communications interface **122** (FIG. 3) may be used to receive commands that control image capture and/or processing in the DICD **100** and may be configured to transfer image data to another device. For example, the communications interface **122** may enable communications with a personal computing device, a smartphone, a tablet computer, a phablet, a smart watch, a portable computer, and/or any other such device or combination of devices that are configured to receive user input and/or communicate information with the DICD **100**. In various implementations of the DICD **100**, it is envisioned that the communications interface **122** may include a wired interface (e.g., a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, or a FireWire interface), or a wireless interface (e.g., a Bluetooth interface, a ZigBee interface, or a Wi-Fi interface).

(31) With continued reference to FIG. 3, the user interface **124** may include one or more buttons or switches to allow a user to control the image capture functions of the DICD **100**, to view or scroll through images, to power the DICD **100** on and off, etc. For example, the user interface **124** may be configured to receive information that identifies a user setting, such as an image resolution setting (e.g., 3840 pixels by 2160 pixels), a frame rate setting (e.g., 60 frames per second), a location setting, and/or a context setting used to indicate an activity, such as mountain biking, in response to user input. The user interface **124** may communicate such settings, or related information, to the DICD **100**, and may display, or otherwise present, content acquired by the DICD **100** (e.g., images or video). For example, the user interface **124** may include a display, such as a viewport into the three-dimensional space represented by the images and/or video captured or generated by the DICD **100**.

(32) It is envisioned that the user interface **124** may also communicate additional information to the DICD **100**, such as metadata, orientation information, etc. For example, the user interface **124** may send information pertaining to a defined coordinate system to the DICD **100** such that the DICD **100** may determine an orientation of the user interface **124** relative to the DICD **100**. Based on the determined orientation, the DICD **100** may identify a portion of the images and/or video captured by the DICD **100** for the DICD **100** to send to the user interface **124** for presentation as the viewport. In some implementations, based on the determined orientation, the DICD **100** may ascertain the location of the user interface **124** and/or the dimensions for viewing of a portion of the images and/or video.

(33) In certain embodiments, the DICD **100** may include one or more hardware or software components that support and facilitate global tone mapping against pixels of a captured image. In such embodiments, it is envisioned that during global tone mapping, such hardware and/or software components may integrate color correction operations. Additionally, although not shown in the figures, the DICD **100** may include one or more other information sources or sensors, such as an inertial measurement unit (IMU), an orientation sensor, an accelerometer, a gyroscope, a magnetometer, a global positioning system (GPS) receiver component, an altimeter, a pressure sensor, a temperature sensor, a heart rate sensor, or any other such unit.

(34) It is also envisioned that the DICD **100** may include one or more applications that may be implemented or executed to manage or control the DICD **100** (e.g., via the user interface **124**). For example, such applications may control the configuration of the DICD **100**, video acquisition, video display, or any other configurable or controllable aspect of the DICD **100**. Via such applications, it is envisioned that the DICD **100** may generate and share (e.g., via a cloud-based or social media service) one or more images and/or video files (e.g., in response to a corresponding input by the user); that the DICD **100** may be remotely controlled; that unprocessed or minimally processed images and/or video may be displayed contemporaneously with capture (e.g., for shot framing); that one or more key moments may be marked contemporaneously with image or video capture using a tag (e.g., in response to user input), and that such tags may be displayed or otherwise presented (e.g., in a camera roll application for location review and/or playback of video); and/or that software and/or hardware on the DICD **100** may be wirelessly controlled (e.g., via a web-based graphical interface accessible by a user for selecting a live or previously recorded video stream).

(35) With reference now to FIGS. **2** and **4-6**, the ISLAs **106A**, **106B** will be discussed. As mentioned above, in certain embodiments of the disclosure, it is envisioned that the ISLAs **106A**, **106B**, and the components thereof, may be identical in configuration (e.g., to reduce the number of different components forming the DICD **100**, manufacturing complexity, cost, etc.). As discussed in further detail below, however, the present disclosure also contemplates non-identical ISLAs. For example, in the embodiment illustrated throughout the figures, the ISLA **106A** includes: a lens **130A** defining an optical axis **XA** (FIG. **1A**); an electronics assembly **132A** (FIGS. **2**, **4-6**); and a mount **134A** that supports the lens **130A** and the electronics assembly **132A**; and the ISLA **106B** includes: a lens **130B** defining an optical axis **XB** (FIG. **1A**); an electronics assembly **132B** (FIGS. **2**, **4-6**); and a mount **134B** that supports the lens **130B** and the electronics assembly **132B**. The ISLAs **106A**, **106B** may also include respective trim rings **136A**, **136B** that are positioned about the lenses **130A**, **130B** to assist in securing the lenses **130A**, **130B** to the mounts **134A**, **134B** and/or increase the overall exterior aesthetic appeal of the ISLAs **106A**, **106B**. It is also envisioned that the trim rings **136A**, **136B** may be configured and positioned to create a water-tight interface with the body **102** (FIGS. **1A**, **1B**) of the DICD **100** (e.g., to facilitate use of the DICD **100** in wet or underwater environments).

(36) To facilitate positioning of the ISLAs **106A**, **106B** within the body **102** of the DICD **100**, the mounts **134A**, **134B** include interfaces **138A**, **138B**, respectively, that correspond in configuration and position to internal engagement structures **140** (FIG. **2**) included on the body **102**. For example, in the illustrated embodiment, the interfaces **138A**, **138B** include a series of flanges **142A**, **142B** defining openings **144A**, **144B** that are configured to receive fasteners **146** (FIG. **4**) (e.g., screws, rivets, clips, or the like). It is envisioned that the fasteners **146** may be used to either fixedly secure the ISLAs **106A**, **106B** to the body **102** of the DICD **100**, or that the fasteners **146** may allow for removal of the ISLAs **106A**, **106B** (e.g., to facilitate the repair or replacement of the ISLA **106A** and/or the ISLA **106B**, or one or more of the components thereof). In addition to facilitating secured engagement between the ISLAs **106A**, **106B** and the body **102** of the DICD **100**, it is envisioned that the interfaces **138A**, **138B** and the engagement structures **140** may act as positional location features to ensure proper orientation of the ISLAs **106A**, **106B** within the body **102** of the DICD **100** and mutual alignment of the ISLAs **106A**, **106B** in multiple (e.g., six) degrees-of-freedom, which is discussed in further detail below.

(37) The ISLAs **106A**, **106B** are positioned in a back-to-back (Janus) configuration in which the ISLAs **106A**, **106B** are generally oriented in opposition. More specifically, the ISLAs **106A**, **106B** are rotated approximately 180° from one another such that the ISLA **106A** faces in a first direction and the ISLA **106B** faces in a second, opposite direction. The respective optical axes **XA**, **XB** of the ISLAs **106A**, **106B** are thus oriented in generally antiparallel relation (e.g., within a tolerance on the order of 1% to 10%, although tolerances outside this range would not be beyond the scope

of the present disclosure).

(38) As seen in FIG. 1A, in the illustrated embodiment, the ISLAs **106A**, **106B** are laterally offset from one another such that the optical axes **XA**, **XB** are separated by a distance **D** (FIG. 1A). By laterally separating the ISLAs **106A**, **106B** and the optical axes **XA**, **XB**, the overall thickness **T** (FIG. 1A) of the DICD **100** may be reduced when compared to designs in which the ISLAs **106A**, **106B** are not offset. For example, the back-to-back offset arrangement of the ISLAs **106A**, **106B** may allow the overall thickness **T** of the DICD **100** to approximate that of a single-lens barrel, as opposed to twice the thickness of a single lens barrel. Although each of the optical axes **XA**, **XB** is illustrated as being offset from a central axis **X** (FIG. 1A) of the DICD **100**, in certain embodiments of the disclosure, it is envisioned that the DICD **100** may be configured such that the optical axis **X** is aligned with the optical axis **XA** or the optical axis **XB**.

(39) The electronics assemblies **132A**, **132B** are positioned behind (e.g., inwardly of) the lenses **130A**, **130B**, respectively, and include image sensors **148A**, **148B** (FIG. 5A, 6), respectively, as well as various other electrical components that support the capture and processing of digital images. For example, the electronics assemblies **132A**, **132B** may further include printed circuit boards **150A**, **150B** supporting the image sensors **148A**, **148B** and protective covers **152A**, **152B** (e.g., glass overlays) for the image sensors **148A**, **148B**, respectively, connectors, wires, etc.

(40) The image sensors **148A**, **148B** are configured to detect light of a certain spectrum (e.g., the visible spectrum or the infrared spectrum), and convey information representative of an image as one or more electrical signals (e.g., analog or digital signals). For example, the image sensors **148A**, **148B** may include CCDs or active pixel sensors in a CMOS, and, in certain embodiments, may include digital-to-analog converters. Suitable examples of the image sensors **148A**, **148B** include, but are not limited to, charge-coupled device (CCD) sensors, active pixel sensors (APS), complementary metal-oxide-semiconductor (CMOS) sensors, and N-type metal-oxide-semiconductor (NMOS) sensors. In certain embodiments, the image sensors **148A**, **148B** may be adapted to capture hyper-hemispherical image planes from light entering the lenses **130A**, **130B** corresponding to the fields-of-view **108A**, **108B** (FIG. 2), respectively, and may be positioned in generally orthogonal relation to the optical axes **XA**, **XB** (FIG. 1A) of their respective lenses **130A**, **130B** (e.g., within a tolerance on the order of 0.5% to 2%, although tolerances outside this range would not be beyond the scope of the present disclosure).

(41) As seen in FIG. 2, the fields-of-view **108A**, **108B** of the ISLAs **106A**, **106B** are respectively defined by boundaries **154A**, **154B**. To facilitate the capture and generation of spherical images, it is envisioned that the fields-of-view **108A**, **108B** may overlap at stitch points **156**, **158**. For example, each of the fields-of-view **108A**, **108B** may lie substantially within the range of approximately 180° to 210°, although larger and smaller values for the field-of-view **108A** and/or the field-of-view **108B** would not be beyond the scope of the present disclosure.

(42) Depending upon the particular configuration and orientation of the ISLAs **106A**, **106B**, one or more blind spots **160**, **162** (or other such areas) may be outside of the respective fields-of-view **108A**, **108B** of the lenses **130A**, **130B**. In the blind spots **160**, **162**, light may be obscured from the lenses **130A**, **130B** and the corresponding image sensors **148A**, **148B** (FIG. 5A, 6), and content in the blind spots **160**, **162** may be omitted from capture. As such, in certain embodiments, the DICD **100** may be configured to reduce (or eliminate) the blind spots **160**, **162** (e.g., by varying the particular configurations and/or orientations of the ISLAs **106A**, **106B** and the components thereof). For example, by reducing the distance **D** (i.e., the lateral offset between the ISLAs **106A**, **106B**) (FIG. 1A), the overlap in the fields-of-view **108A**, **108B** may be improved.

(43) A small change in the alignment of the lenses **130A**, **130B** and/or the image sensors **148A**, **148B** (e.g., tilting) may change the relative positions of the fields-of-view **108A**, **108B**, the locations of the stitch points **156**, **158**, and the sizes of the blind spots **160**, **162**, and may impact image quality. Additionally, incomplete or inaccurate information regarding the alignment of the ISLAs **106A**, **106B** (e.g., the locations of the stitch points **156**, **158**) may decrease the quality

and/or accuracy of combined (stitched) images and may also impact the efficiency with which combined images are generated. In some implementations, it is envisioned that the DICD **100** may maintain information indicating the location and orientation of the lenses **130A**, **130B** and/or the image sensors **148A**, **148B** such that the fields-of-view **108A**, **108B** and/or the stitch points **156**, **158** may be accurately determined (e.g., to improve the accuracy of combined images and/or the efficiency with which combined images are generated).

(44) With reference now to FIGS. **4-12** in particular, the bridge member **110** and connection of the ISLAs **106A**, **106B** will be discussed. The bridge member **110** is configured as a discrete (separate) structure that is connectable to (e.g., engageable with) the mounts **134A**, **134B** to fixedly connect the ISLAs **106A**, **106B**. For example, it is envisioned that the bridge member **110** may be configured to mechanically engage the mounts **134A**, **134B** in a clearance or interference, in a snap-fit arrangement, or through the employ of one or more mechanical connectors (e.g., screws, detents, clips, or other such fasteners). Additionally, or alternatively, as illustrated throughout the figures, the bridge member **110** may be secured to the mounts **134A**, **134B** using an adherent **164** (e.g., an adhesive), which may be positioned internally (e.g., between the mounts **134A**, **134B** and the bridge member **110**, as seen in FIG. **5A**) and/or externally (e.g., at the interface between the mounts **134A**, **134B**, as seen in FIG. **5B**).

(45) In certain embodiments, it is envisioned that the bridge member **110** and the mounts **134A**, **134B** may include (e.g., may be formed from) the same material or combination of materials such that the coefficients of thermal expansion for the bridge member **110** and the mounts **134A**, **134B** are either identical or substantially similar. For example, in one embodiment of the disclosure, the bridge member **110** and the mounts **134A**, **134B** may each be formed from liquid crystal polymer. Consistency between the material(s) of construction used in the bridge member **110** and the mounts **134A**, **134B** facilitates uniformity in the expansion and contraction of the bridge member **110** and the mounts **134A**, **134B** (e.g., upon exposure of the DICD **100** to different temperatures and/or during temperature changes within the DICD **100**) to maintain the connection between the bridge member **110** and the mounts **134A**, **134B**, and thus, the relative positions and alignment of the ISLAs **106A**, **106B**, which is discussed in further detail below.

(46) The bridge member **110** includes opposite ends **110A**, **110B** (FIGS. **4**, **7**, **8**), and defines a longitudinal axis XBM (FIGS. **4**, **7**, **8**) that extends there between in generally parallel relation to the optical axes XA, XB. The bridge member **110** includes a first bridge portion **166A** that faces the mount **134A** and a second bridge portion **166B** that faces the mount **134B**. In alternate embodiments of the disclosure, it is envisioned that the bridge portions **166A**, **166B** may be discrete structures that are mechanically and/or chemically secured together (e.g., via one or more fasteners, adhesives, etc.), or, alternatively, that the bridge portions **166A**, **166B** may be integrally (e.g., monolithically) formed, as shown throughout the figures, such as through injection molding, 3-D printing, etc.

(47) The bridge portions **166A**, **166B** are identical in configuration and are positioned in generally opposite orientations (i.e., the bridge portions **166A**, **166B** are rotated approximately 180° from one another and face in opposite directions). More specifically, the bridge portion **166A** defines opposite side walls **168A**, **170A** (FIGS. **7**, **8**) and opposite end walls **172A**, **174A**, and includes a first end section **176A**, an opposite second end section **178A**, and an intermediate section **180A** that is positioned between the end sections **176A**, **178A**. Correspondingly, the bridge portion **166B** defines opposite side walls **168B**, **170B** and opposite end walls **172B**, **174B**, and includes a first end section **176B**, an opposite second end section **178B**, and an intermediate section **180B** that is positioned between the end sections **176B**, **178B**. While the end walls **172A**, **172B** are illustrated as extending transversely in relation to the optical axes XA, XB (FIG. **4**) (i.e., as extending along axes that intersect the optical axes XA, XB so as to subtend acute angles therewith) and the end walls **174A**, **174B** are illustrated as extending in generally orthogonal relation to the optical axes XA, XB in the particular embodiment of the bridge member **110** shown throughout the figures, in alternate

embodiments of the disclosure, the configurations of the end walls **172A**, **174A** and the end walls **172B**, **174B** may be varied. For example, it is envisioned that the end walls **172A**, **172B** may also extend in generally orthogonal relation to the optical axes **XA**, **XB** such that the end walls **172A**, **174A** extend in generally parallel relation to the end walls **172B**, **174B**, respectively.

(48) As seen in FIGS. **7** and **8**, for example, the first end section **176A** of the bridge portion **166A** is positioned generally opposite the second end section **178B** of the bridge portion **166B** so as to define a first node **182i**, the second end section **178A** of the bridge portion **166A** is positioned generally opposite the first end section **176B** of the bridge portion **166B** so as to define a second node **182ii**, and the intermediate section **180A** of the bridge portion **166A** is positioned generally opposite the intermediate section **180B** of the bridge portion **166B** so as to define a third node **182iii**. The first node **182i** defines opposite end faces **184i**, **186i**, the second node **182ii** defines opposite end faces **184ii**, **186ii**, and the third node **182iii** defines opposite end faces **184iii**, **186iii**. Whereas the end faces **184i**, **186i** of the first node **182i** and the end faces **184ii**, **186ii** of the second node **182ii** are each generally planar in configuration and extend in generally parallel relation to the optical axes **XA**, **XB**, the end faces **184iii**, **186iii** are non-planar in configuration. More specifically, the end face **184iii** of the third node **182iii** includes respective first and second segments **188**, **190**, and the end face **186iii** of the third node **182iii** includes respective first and second segments **192**, **194**. The third node **182iii** is configured such that the segments **188**, **192** extend in generally parallel relation to the optical axes **XA**, **XB** (FIG. **4**), and the segments **190**, **194** extend transversely in relation to the optical axes **XA**, **XB** (i.e., the segments **190**, **194** extend along axes that intersect the optical axes **XA**, **XB** so as to subtend acute angles therewith).

(49) Depending on the particular configuration and/or orientation of the mounts **134A**, **134B**, for example, it is envisioned that the particular configurations of the bridge portions **166A**, **166B** (e.g., the nodes **182i-182iii**) may be varied in alternate embodiments of the disclosure. For example, it is envisioned that the end faces **184i-184iii**, **186i-186iii** may include non-linear (e.g., arcuate) portions, that the number and/or specific locations of the nodes **182i-182iii** may be varied (e.g., the bridge member **110** may include a single node **182** only, or a pair of nodes **182i**, **182ii**), etc. Additionally, while the bridge portions **166A**, **166B** are illustrated as being generally aligned throughout the figures such that the side walls **168A**, **168B** and the side walls **170A**, **170B** are generally coplanar, in alternate embodiments of the disclosure, it is envisioned that the bridge portions **166A**, **166B** may be offset (i.e., along an axis extending in orthogonal relation to the optical axes **XA**, **XB** and parallel to the respective front and rear faces **102A**, **102B** (FIG. **1A**) of the body **102** of the DICD **100**) such that the side walls **168A**, **168B** and the side walls **170A**, **170B** are oriented in different planes.

(50) As seen in FIGS. **9-12**, the mounts **134A**, **134B** respectively include cradles **196A**, **196B** that are configured to receive the bridge member **110**. More specifically, the cradles **196A**, **196B** include receptacles **198A**, **198B** that collectively define a cavity **200** that is configured to receive the bridge member **110** (FIGS. **5A**, **7**, **8**) such that the bridge portion **166A** is positioned within the receptacle **198A** and the bridge portion **166B** is positioned within the receptacle **198B**. The cradle **196A** includes: a pair of side walls **202A**, **204A**; a pair of end walls **206A**, **208A**; and a base wall **210A**; and the cradle **196B** includes: a pair of side walls **202B**, **204B**; a pair of end walls **206B**, **208B**; and a base wall **210B**. As seen in FIG. **5A**, the base walls **210A**, **210B** include offsets **216A**, **216B**, respectively, resulting in non-linear, interrupted configurations that accommodate the electronics assemblies **132A**, **132B** (e.g., the image sensors **148A**, **148B**), respectively. In alternate embodiments of the disclosure, however, it is envisioned that the offsets **216A**, **216B** may be omitted such that the base walls **210A**, **210B** are linear (and continuous) in configuration.

(51) The cradles **196A**, **196B** also include a series of dividers **212A**, **212B** that are positioned between the end walls **206A**, **208A** and **206B**, **208B**. The dividers **212A**, **212B** extend inwardly (i.e., towards the optical axes **XB**, **XA** (FIG. **5A**), respectively) and are configured and positioned to define a series of individual chambers **214A**, **214B** within the receptacles **198A**, **198B**. In the

embodiment seen in FIGS. 9 and 11, for example, the cradle **196A** includes a first divider **212A1** and a second divider **212A2**, and the cradle **196B** includes a first divider **212B1** and a second divider **212B2**.

(52) The dividers **212A1**, **212A2** included in the first cradle **196A** are positioned such that a first end chamber **214A1** is defined between the end wall **206A** and the first divider **212A1**, a second end chamber **214A2** is defined between the divider **212A2** and the end wall **208A**, and an intermediate chamber **214A3** is defined between the dividers **212A1**, **212A2**.

(53) Correspondingly, the dividers **212B1**, **212B2** included in the second cradle **196B** are positioned such that a first end chamber **214B1** is defined between the end wall **206B** and the first divider **212B1**, a second end chamber **214B2** is defined between the divider **212B2** and the end wall **208B**, and an intermediate chamber **214B3** is defined between the dividers **212B1**, **212B2**. Although illustrated as including two dividers **212** and three chambers **214** in the embodiment shown throughout the figures, it should be appreciated that the number and particular locations of the dividers **212** and the chambers **214** may be varied in alternate embodiments of the disclosure (e.g., depending upon the particular configuration of the bridge member **110**). For example, in certain embodiments, it is envisioned that the dividers **212** may be eliminated altogether such that the receptacles **198A**, **198B** respectively include single chambers **214A**, **214B** only.

(54) The ISLAs **106A**, **106B** are oriented such that the first end chamber **214A1** is positioned generally opposite the second end chamber **214B2** so as to define a first cavity portion **218i** (FIG. 5A), the second end chamber **214A2** is positioned generally opposite the first end chamber **214B1** so as to define a second cavity portion **218ii**, and the intermediate chamber **214A3** is positioned generally opposite the intermediate chamber **214B3** so as to define a third cavity portion **218iii**. Upon assembly of the optical module **104**, the bridge member **110** is positioned between the ISLAs **106A**, **106B** such that the first node **182i** (FIGS. 7, 8) is positioned within the first cavity portion **218i** defined by the first end chamber **214A1** and the second end chamber **214B2**, the second node **182ii** is positioned within the second cavity portion **218ii** defined by the second end chamber **214A2** and the first end chamber **214B1**, and the third node **182iii** is positioned within the third cavity portion **218iii** defined by the intermediate chamber **214A3** and the intermediate chamber **214B3**.

(55) In the illustrated embodiment, as indicated above, the bridge member **110** is secured to the mounts **134A**, **134B** using the aforementioned adherent **164**, which may partially or entirely surround the bridge member **110**, as illustrated in FIG. 5A, and may be positioned at the interface between the mounts **134A**, **134B**, as illustrated in FIG. 5B. For example, as shown throughout the figures, the adherent **164** may be positioned between the base wall **210A** (FIGS. 9-11) of the cradle **196A** and the end faces **184i-184iii** (FIG. 7, 8) defined by the nodes **182i-182iii**, respectively, and between the base wall **210B** of the cradle **196B** and the end faces **186i-186iii** defined by the nodes **182i-182iii**, respectively. To further secure the bridge member **110** to the mounts **134A**, **134B**, it is envisioned that the adherent **164** may also be positioned between the side walls **202A**, **204A** (FIGS. 9-11) of the cradle **196A** and the side walls **168A**, **170A** (FIGS. 7, 8) of the bridge portion **166A**, and between the side walls **202B**, **204B** of the cradle **196B** and the side walls **168B**, **170B** of the bridge portion **166B**, as well as between the end walls **206A**, **208A** (FIGS. 9-11) of the cradle **196A** and the end walls **172A**, **174A** (FIGS. 7, 8) of the bridge portion **166A**, and between the end walls **206B**, **208B** of the cradle **196B** and the end walls **172B**, **174B** of the bridge portion **166B**. The various walls **168**, **170**, **172**, **174** and faces **184**, **186** defined by the bridge member **110** increase the available, exposed surface area of the bridge member **110** for contact with the adherent **164** to increase the stability and/or permanence of the connection between the bridge member **110** and the ISLAs **106A**, **106B**.

(56) In certain alternate embodiments of the disclosure, it is envisioned that the bridge member **110** may be eliminated (e.g., to save weight and reduce manufacturing complexity). In such embodiments, the mount **134A** of the ISLA **106A** and the mount **134B** of the ISLA **106B** may be

secured directly to one another using the adherent **164**.

(57) With reference now to FIGS. **1A-12** and FIG. **13**, a method **1300** (FIG. **13**) of assembling the optical module **104** will be discussed. Initially, at steps **1302** and **1304**, the ISLA **106A** (FIGS. **2, 4, 5A, 5B**) and the ISLA **106B** are subjected to individual active alignment processes to achieve optimal alignment between the lens **130A** and the image sensor **148A** and between the lens **130B** and the image sensor **148B**, and thereby facilitate the capture and generation of focused, high-quality images. More specifically, the positions of the lens **130A** and/or the image sensor **148A** are continually adjusted (e.g., in six degrees-of-freedom) until the optical axis **XA** (FIG. **1A**) is centered with respect to the image sensor **148A**, which defines a plane that is generally orthogonal in relation to the optical axis **XA**, as can be appreciated through reference to FIG. **5A**, and the positions of the lens **130B** and/or the image sensor **148B** are continually adjusted (e.g., in six degrees-of-freedom) until the optical axis **XB** is centered with respect to the image sensor **148B**, which defines a separate plane that is generally orthogonal in relation to the optical axis **XB**, as can also be appreciated through reference to FIG. **5A**. During the alignment process, as the lenses **130A, 130B** and/or the image sensors **148A, 148B** are repositioned, optical targets are viewed, and the modulation transfer function (MTF) at each target is monitored until each of the MTF values are within acceptable tolerances. Thereafter, the lenses **130A, 130B** and the image sensors **148A, 148B** are secured in place to maintain proper alignment. For example, the lenses **130A, 130B** and the image sensors **148A, 148B** may be secured by curing an adherent, such as an adhesive (e.g., using a UV light source and/or a heat source).

(58) After actively aligning each of the ISLAs **106A, 106B**, at step **1306**, the ISLAs **106A, 106B** are allowed to cool, and at step **1308** the ISLAs **106A, 106B** and the bridge member **110** are cleaned (e.g., via a plasma cleaning process). The ISLAs **106A, 106B** are then connected to form the optical module **104** by connecting the bridge member **110** (FIGS. **2, 4, 5A, 7, 8**) to the first ISLA **106A** using a first apparatus (e.g., a first robotic assembly) at steps **1310-1314**, and, thereafter, connecting the second ISLA **106B** to the assembly of the first ISLA **106A** and the bridge member **110** using a second apparatus (e.g., a second robotic assembly) at steps **1320-1326**. More specifically, the adherent **164** is applied to the bridge member **110** and/or the cradle **196A** (e.g., internally and/or externally), and the first apparatus is used to orient the bridge member **110** and the cradle **196A** such that the nodes **182i, 182ii, 182iii** (FIG. **7, 8**) are respectively positioned within the chambers **214A1, 214A2, 214A3** (FIG. **9**) defined by the receptacle **198A**. During positioning of the bridge member **110**, at step **1310**, the first apparatus identifies and tracks the positions of the ISLA **106A** and the bridge member **110**, and views image targets through the ISLA **106A**, using the coordinates of the ISLA **106A** and the bridge member **110**. The first apparatus continually adjusts and measures the relative positioning and alignment of the ISLA **106A** and the bridge member **110** until the measurements fall within a predetermined threshold. Once proper alignment between the ISLA **106A** and the bridge member **110** has been achieved, at step **1312**, the assembly of the first ISLA **106A** and the bridge member **110** is exposed to a UV light source to cure the adherent **164** (FIG. **5A**). During UV curing, any adherent **164** applied to any of the exterior surfaces is hardened, and UV light passes through one or more openings **220A** (FIG. **5A**) (e.g., gaps) formed in the mount **134A** to cure any adherent **164** positioned internally between the mount **134A** and the bridge member **110**. Thereafter, at step **1314**, the assembly of the ISLA **106A** and the bridge member **110** is placed into a thermal (heating) apparatus to further cure the adherent **164** and fix the connection between the ISLA **106A** and the bridge member **110**. Following thermal curing, the assembly of the ISLA **106A** and the bridge member **110** is allowed to cool at step **1316**, and the assembly of the ISLA **106A** and the bridge member **110** is cleaned (e.g., using the aforementioned plasma cleaning process) at step **1318**.

(59) The ISLA **106B** is then connected to the bridge member **110** using a similar procedure. More specifically, the adherent **164** is applied to the bridge member **110** and/or the cradle **196B** (e.g., internally and/or externally), and the second apparatus is used to orient the bridge member **110**

(which is connected to the ISLA **106A**) and the cradle **196B** such that the nodes **182i**, **182ii**, **182iii** (FIG. **7**, **8**) are respectively received by the chambers **214B2**, **214B1**, **214B3** (FIG. **11**) defined by the receptacle **198B**. During positioning of the bridge member **110**, at step **1320**, the second apparatus identifies and tracks the positions of the ISLA **106B** and the bridge member **110**, and views image targets through the ISLA **106B**, using the coordinates of the ISLA **106B** and the bridge member **110**. The second apparatus continually adjusts and measures the relative positioning and alignment of the ISLA **106B** and the bridge member **110** until the measurements fall within a predetermined threshold.

(60) Once proper alignment between the ISLA **106A** and the bridge member **110** has been achieved, prior to curing the adherent **164**, at step **1322**, the ISLAs **106A**, **106B** are actively aligned using a bore-sight alignment process similar to that discussed above with respect to the ISLAs **106A**, **106B** individually. The bore-sight alignment process achieves optimal alignment and orientation of the ISLAs **106A**, **106B** by continually adjusting the relative positioning between the ISLA **106A** and the ISLA **106B** (e.g., in six degrees-of-freedom) until the optical axes **XA**, **XB** (FIG. **1A**) are oriented as desired (e.g., in anti-parallel relation). During the bore-sight alignment process, one or more optical targets are viewed through the ISLAs **106A**, **106B** as the relative repositioning of the ISLAs **106A**, **106B** is adjusted, and centering target values are examined and observed until the centering target values are within acceptable tolerances.

(61) Once the ISLAs **106A**, **106B** have been properly aligned, the assembled optical module **104** is exposed to a UV light source at step **1324** to cure the adherent **164** (FIG. **5A**). During UV curing, any adherent **164** applied to any of the exterior surfaces is hardened, and UV light passes through one or more openings **220B** (FIG. **5A**) (e.g., gaps) formed in the mount **134B** to cure any adherent **164** positioned internally between the mount **134B** and the bridge member **110**. Thereafter, at step **1326**, the optical module **104** is placed into a thermal (heating) apparatus to further cure the adherent **164** and fix the connection between the ISLA **106B** and the bridge member **110**, and the assembled optical module **104** is allowed to cool at step **1328**.

(62) Through optimal alignment of the ISLAs **106A**, **106B**, greater overlap in the fields-of-view **108A**, **108B** (FIG. **2**) can be achieved, which allows for an increase in the quality of the combined images generated through image stitching, as well as overall image processing and resolution. Additionally, the active (bore-sight) alignment process described herein reduces the number of degrees of field of view angle that would otherwise be required by known additional procedures, such as image stabilization, thereby facilitating the creation of higher-resolution images.

(63) With reference now to FIGS. **14-16B**, alternate embodiments of the ISLAs **106A**, **106B** (which are identified by the reference characters **306A**, **306B**) will be discussed. The ISLAs **306A**, **306B** are generally similar to the ISLAs **106A**, **106B** discussed above, respectively, and, accordingly, in the interest of brevity, will be discussed only with respect to any differences therefrom.

(64) The ISLAs **306A**, **306B** respectively include mounts **334A**, **334B** with cradles **396A**, **396B**. Whereas the ISLAs **106A**, **106B** are configured for use with the bridge member **110**, the ISLAs **306A**, **306B** are configured for direct connection so as to obviate the need for the bridge member **110**. More specifically, the cradle **396A** includes: a pair of side walls **402A**, **404A** (FIG. **15A**); a pair of end walls **406A**, **408A**; and a base wall **410A** (FIG. **14**); and the cradle **396B** includes: a pair of side walls **402B**, **404B** (FIG. **16A**); a pair of end walls **406B**, **408B**; and a base wall **410B** (FIG. **14**). As seen in FIG. **14**, the base walls **410A**, **410B** include offsets **416A**, **416B**, respectively, resulting in non-linear, interrupted configurations that accommodate the electronics assemblies **432A**, **432B**, respectively. In alternate embodiments of the disclosure, however, it is envisioned that the offsets **416A**, **416B** may be omitted such that the base walls **410A**, **410B** are linear (and continuous) in configuration.

(65) The cradle **396A** further include a series of dividers **412A** that are positioned between the end walls **406A**, **408A**, and the cradle **396B** further includes a series of corresponding dividers **412B** that are positioned between the end walls **406B**, **408B**. As seen in FIG. **14**, each of the dividers

412A, **412B** extends inwardly (i.e., towards the optical axes **XB**, **XA**, respectively). In the particular embodiment illustrated in FIGS. **14-16B**, the cradle **396A** includes a first divider **412A1**, a second divider **412A2**, a third divider **412A3**, and a fourth divider **412A4**, and the cradle **396B** includes a first divider **412B1**, a second divider **412B2**, a third divider **412B3**, and a fourth divider **412B4**. It should be appreciated, however, that the number of dividers **412** included on the ISLAs **306A**, **306B** may be varied in alternate embodiments of the disclosure.

(66) With particular reference to FIG. **14**, the dividers **412A1**, **412A2**, **412A3**, **412A4** are positioned at respective (first, second, third, and fourth) heights **H1A**, **H2A**, **H3A**, **H4A** (measured from a base/lowest point of the mount **334A**), and the dividers **412B1**, **412B2**, **412B3**, **412B4** are positioned at respective (first, second, third, and fourth) heights **H1B**, **H2B**, **H3B**, **H4B** (measured from a base/lowest point of the mount **334B**). To facilitate proper mating engagement of the mounts **334A**, **334B**, at least one of the dividers **412A** is positioned in a different longitudinal (vertical) position than at least one of the corresponding dividers **412B** such that at least one of the heights **H1A**, **H2A**, **H3A**, **H4A** differs from at least one of the heights **H1B**, **H2B**, **H3B**, **H4B**. The dividers **412A** are, thus, at least partially offset from the dividers **412B**, which results in the dissimilar configurations of the ISLAs **306A**, **306B** mentioned above.

(67) With continued reference to FIG. **14**, the ISLAs **306A**, **306B** are oriented such that: the end wall **406A** is positioned adjacent the end wall **408B** at an interface **Ii**; the end wall **406B** is positioned adjacent the end wall **408A** at an interface **Iii**; the divider **412A1** is positioned adjacent the divider **412B4** at an interface **Iiii**; the divider **412A2** is positioned adjacent the divider **412B3** at an interface **Iiv**; the divider **412A3** is positioned adjacent the divider **412B2** at an interface **Iv**; and the divider **412A4** is positioned adjacent the divider **412B1** at an interface **Ivi**. To secure the mounts **334A**, **334B** together, the aforementioned adherent **164** is placed at the interfaces **Ii-Ivi** for curing using the two-stage (i.e., UV and thermal) processes and procedure discussed above. By providing for a multitude of interfaces **I** (e.g., via the inclusion of the various dividers **412**), the surface area of the mounts **334A**, **334B** available for contact with the adherent **164** is increased to thereby increase the stability and/or permanence of the connection between the ISLAs **306A**, **306B**. As discussed above in connection with FIG. **5B** (for example), it is envisioned that the adherent **164** may be positioned both internally (i.e., within the mounts **334A**, **334B**), as well as externally, to further enhance the connection between the ISLAs **306A**, **306B**.

(68) Persons skilled in the art will understand that the various embodiments of the disclosure described herein, and shown in the accompanying figures, constitute non-limiting examples, and that additional components and features may be added to any of the embodiments discussed hereinabove without departing from the scope of the present disclosure. Additionally, persons skilled in the art will understand that the elements and features shown or described in connection with one embodiment may be combined with those of another embodiment without departing from the scope of the present disclosure and will appreciate further features and advantages of the presently disclosed subject matter based on the description provided. Variations, combinations, and/or modifications to any of the embodiments and/or features of the embodiments described herein are also within the abilities of a person having ordinary skill in the art, and thus, are also within the scope of the disclosure, as are alternative embodiments that may result from combining, integrating, and/or omitting features from any of the disclosed embodiments.

(69) Use of the term “optionally” with respect to any element of a claim means that the element may be included or omitted, with both alternatives being within the scope of the claim. Additionally, use of broader terms such as “comprises,” “includes,” and “having” should be understood to provide support for narrower terms such as “consisting of,” “consisting essentially of,” and “comprised substantially of.” Accordingly, the scope of protection is not limited by the description set out above, but is defined by the claims that follow, and includes all equivalents of the subject matter of the claims.

(70) In the preceding description, reference may be made to the spatial relationships between the

various structures illustrated in the accompanying drawings, and to the spatial orientation of the structures. However, as will be recognized by those skilled in the art after a complete reading of this disclosure, the structures described herein may be positioned and oriented in any manner suitable for their intended purpose. Thus, the use of terms such as “above,” “below,” “upper,” “lower,” “inner,” “outer,” “upward,” “downward,” “inward,” “outward,” etc., should be understood to describe a relative relationship between structures and/or a spatial orientation of the structures. Those skilled in the art will also recognize that the use of such terms may be provided in the context of the illustrations provided by the corresponding figure(s).

(71) Additionally, terms such as “approximately,” “generally,” “substantially,” and the like should be understood to allow for variations in any numerical range or concept with which they are associated. For example, it is intended that the use of terms such as “approximately” and “generally” should be understood to encompass variations on the order of 25%, or to allow for manufacturing tolerances and/or deviations in design.

(72) Although terms such as “first,” “second,” etc., may be used herein to describe various steps, operations, elements, components, regions, and/or sections, these steps, operations, elements, components, regions, and/or sections should not be limited by the use of these terms in that these terms are used to distinguish one step, operation, element, component, region, or section from another. Thus, unless expressly stated otherwise, a first step, operations, element, component, region, or section could be termed a second step, operation, element, component, region, or section without departing from the scope of the present disclosure.

(73) Each and every claim is incorporated as further disclosure into the specification and represents embodiments of the present disclosure. Also, the phrases “at least one of A, B, and C” and “A and/or B and/or C” should each be interpreted to include only A, only B, only C, or any combination of A, B, and C.

Claims

1. An optical module for a digital image capturing device (DICD), the optical module comprising: a first integrated sensor-lens assembly (ISLA) including: a first lens; a first image sensor fixed in relation to the first lens; and a first mount configured to support the first lens and the first image sensor, wherein the first mount includes a first exterior receptacle; a second ISLA including: a second lens; a second image sensor fixed in relation to the second lens; and a second mount configured to support the second lens and the second image sensor, wherein the second mount includes a second exterior receptacle oriented in facing relation to the first exterior receptacle such that the first exterior receptacle and the second exterior receptacle collectively define a cavity; and a bridge member configured for receipt within the cavity to thereby fixedly secure together the first and second ISLAs, wherein the bridge member is configured as a discrete structure separate from the first and second ISLAs.
2. The optical module of claim 1, wherein the first ISLA and the second ISLA are oriented in generally opposite directions.
3. The optical module of claim 2, wherein the first ISLA and the second ISLA are identical in configuration.
4. The optical module of claim 2, wherein the first ISLA defines a first field-of-view of approximately 180° to approximately 210°, and the second ISLA defines a second field-of-view of approximately 180° to approximately 210°.
5. The optical module of claim 4, wherein the first ISLA and the second ISLAs are positioned such that the first field-of-view and the second field-of-view overlap, whereby images captured by the first ISLA and the second ISLA can be stitched together to form a single image.
6. The optical module of claim 1, wherein the first ISLA defines a first optical axis, and the second ISLA defines a second optical axis.

7. The optical module of claim 6, wherein the second optical axis is offset from the first optical axis.
 8. The optical module of claim 1, wherein the bridge member includes a first bridge portion and a second bridge portion connected to the first bridge portion.
 9. The optical module of claim 8, wherein the first bridge portion and the second bridge portion are identical in configuration.
 10. The optical module of claim 9, wherein the first bridge portion and the second bridge portion are oriented in generally opposite directions.
 11. A digital image capturing device (DICD) comprising: a first integrated sensor-lens assembly (ISLA) facing in a first direction and defining a first optical axis, wherein the first ISLA includes a first mount having a first body portion; a second ISLA facing in a second direction generally opposite to the first direction and defining a second optical axis, wherein the second ISLA includes a second mount having a second body portion; and a bridge member positioned between the first ISLA and the second ISLA to fixedly secure together the first ISLA and the second ISLA, wherein the bridge member is configured as a discrete structure separate from the first body portion and the second body portion.
 12. The DICD of claim 11, wherein the second optical axis is offset from the first optical axis.
 13. The DICD of claim 11, wherein the first ISLA and the second ISLA are identical in configuration.
 14. The DICD of claim 11, wherein the first mount includes a first exterior receptacle, and the second mount includes a second exterior receptacle oriented in facing relation to the first exterior receptacle such that the first exterior receptacle and the second exterior receptacle collectively define a cavity receiving the bridge member.
 15. The DICD of claim 14, wherein the bridge member defines a longitudinal axis that is generally parallel in relation to the first optical axis and the second optical axis.
 16. A method of assembling an optical module for a digital image capturing device (DICD), the method comprising: providing a first integrated sensor-lens assembly (ISLA) including a first mount having a first body portion; providing a second ISLA including a second mount having a second body portion; providing a bridge member configured as a discrete structure separate from the first body portion and the second body portion; connecting the bridge member to the first ISLA; and connecting the second ISLA to the bridge member.
 17. The method of claim 16, wherein connecting the second ISLA to the bridge member includes orienting the first ISLA and the second ISLA in generally opposite directions.
 18. The method of claim 17, wherein connecting the second ISLA to the bridge member includes orienting the first ISLA and the second ISLA such that optical axes defined by the first ISLA and the second ISLA are laterally offset from one another.
 19. The method of claim 17, wherein providing the first ISLA includes actively aligning a lens and an image sensor of the first ISLA, and wherein providing the second ISLA includes active aligning a lens and an image sensor of the second ISLA.
 20. The method of claim 19, further comprising: actively aligning the first ISLA and the second ISLA with respect to one another.
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